



Project Status report

Name: **Muhammad Hamza Shahab, Syed Haider Abbas Naqvi**

Community & UN SDG(s): **SaskTel network engineers and architects**

UN SDG(s):

- SDG#7: Affordable and clean energy
- SDG#11: Sustainable cities and communities
- SDG#12: Responsible consumption and production
- SDG#13: Climate action

MVP # **2 & 3**

Sprint cycle dates: **March 11th, 2025 - March 24th, 2025**

Project Name	Eco-Resilient Networks: Smart Deployment for the Future
Blurb	We are developing a simulation framework to optimize the deployment of Service Function Chains (SFCs) for 5G video streaming in edge-cloud environments. Our focus is on <i>jointly</i> maximizing network availability and minimizing the carbon footprint, addressing UN Sustainable Development Goals 7, 9, 11, 12, and 13. The project provides network engineers (specifically at SaskTel) with tools for data-driven decisions, leading to more efficient and sustainable network operations.
For Week Ending	March 24, 2025
Project Status	Green
Status Description	The project continues to progress well. We have completed extensive simulations comparing the three embedding policies (AAE, CAE, TAE) with the PSO algorithm. We have begun initial implementation of the Simulated Annealing (SA) and Genetic Algorithm (GA). Significant progress has been made on documentation, including the user guide and the Results and Discussion sections of the research paper (MVP 3).
Activities—During the past sprint cycle	

During the past sprint cycle, we achieved the following:

- **Completed Work Package 2.4:** Algorithm Comparison and Evaluation.
 - Ran extensive simulations with varying network topologies, SFC requests, and weighting factors for the TAE policy.
 - Analysed the results and generated comparison reports, showing the trade-offs between availability, carbon footprint, and latency for different policies and configurations.
 - Developed visualizations (graphs, charts) to effectively communicate the results.
- **Began Work Package 2.5:** Other Redundancy Optimization Algorithms.
 - Started implementation of the Simulated Annealing (SA) algorithm.
 - Started implementation of the Genetic Algorithm (GA).
 - Made sure that these algorithms integrate with the project structure.
- **Work Package 3.1:** Code Documentation.
 - Continued documenting code (ongoing).
- **Work Package 3.2:** User Guide.
 - Drafted initial sections of the user guide, covering installation, configuration, and running basic simulations.
- **Work Package 3.3:** Research Paper (Draft).
 - Completed a substantial portion of the Results section, including detailed analysis and visualizations.



- Began outlining the Discussion and Conclusion sections.
- Prepared presentation for Scrum #3.
- **Code/Demo:**
 - Demonstrate the simulation running with different embedding policies (AAE, CAE, TAE) and PSO.
 - Show the comparison reports and visualizations, highlighting the trade-offs between the policies.
 - Show the code structure along with the SA and GA implementation.
 - Show progress on documentation and the structure of the paper.

Project Issues

No significant issues have impeded overall progress. We maintain a proactive approach to issue management, ensuring no roadblocks impact our progress.

Project Changes

No formal project changes have been requested or approved during this sprint cycle. The project scope and requirements, as defined in the Project Scope Statement, remain unchanged.

Activities—Planned for Next Week

Following is the list of all activities that the team members will work on during the next week of the project:

- **Continue/Complete Work Package 2.5:** Continue implementation of SA and GA and integrate them with the embedding policies.
- **Work Package 3.1:** Code Documentation (ongoing).
- **Work Package 3.2:** User Guide (continue drafting, focusing on how to interpret results and customize parameters).
- **Work Package 3.3:** Research Paper (Draft): Continue writing the Discussion and Conclusion sections. Refine the Results section based on feedback.

Reflection

Do you feel "on track"?

Yes, the project is on track. We've made excellent progress on comparing the initial algorithms and are making headway on the additional algorithms and documentation.

What progress do you particularly feel good (great) about?

The comprehensive comparison of embedding policies with PSO, the clear visualizations of the results, and the progress on the research paper are particularly satisfying.

What barriers (if any) do you feel is/are a current impediment to success?

No significant barriers have been encountered so far.

What help (if any) do you require to move positively forward?

Feedback on the clarity and completeness of the user guide draft and the structure/content of the research paper would be valuable.

What questions or concerns do you have (if any)?

We are prioritizing completing the core functionality and documentation. We are seeking guidance on the best way to present the comparison results in the paper (e.g., specific types of graphs, statistical analysis).